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Peak-dBA noise model — methodology & measurement record

The single-event peak-dBA model behind the sensitivity map, the measurements it rests on, and the technical points most often misread. Written so another analyst can reproduce every figure.

Prepared 2026-10-08; revised 2026-10-19 — added the measurement basis, the 2,889-flight turn-altitude record, the five-flight calibrated cross-check, the receptor characterization, equipment notes, and a plain-language explanation of the law and of common misconceptions. The sensitivity-map figures are unchanged from the 2026-10-10 model.

WHAT THIS RESTS ON

A measurement record, not assumptions

TrueNoise is a continuous measurement record. Between 1 May and 15 August 2026 the monitor logged roughly 30,000 departure samples across 153 days, grouped into about 3,185 departure passes. From that record:

- **2,889 clean Runway 15R departures** were analyzed for turn altitude — the result held steady every month (median 2,325 ft in May, 2,375 in June, 2,292 in July, 2,196 in August), so it reflects a full season, not a snapshot.
- **249 isolated overflights** calibrated the loudness-vs-distance law, which then reproduced an independent calibration point (73.3 dBA at 0.55 mi) and the monitor's own measured level (70.5 dBA) without being tuned to either.
- A **five-flight blind test against a calibrated sound-level meter** put the model within about 0.8 dB of ground truth — slightly better than a commercial noise-tracking device measured in the same window.

Each claim below is tied to its own sample: the altitude finding rests on ~2,889 flights, the loudness law on 249 measured overflights plus three validations, the geometry on the ADS-B track set. The large altitude

sample and the smaller law sample reinforce each other — the altitude the model assumes is independently confirmed on nearly 3,000 flights.

THE METRIC

What we measure — and why it is not DNL

Single-event peak dBA — what TrueNoise measures — is the loudest instant of one overflight: the spike you actually hear when a plane passes. It is an event, measured directly by the monitor.

DNL (Day-Night average sound Level) is a 24-hour energy average of all noise, quiet hours included, with a 10-dB penalty on night events. It is a bookkeeping figure, not a sound anyone hears; the “65” in federal land-use guidance is a DNL contour.

These are different physical quantities on different scales. A corridor can sit far below DNL 65 as a 24-hour average and still deliver single-event peaks in the low 70s dozens of times a day — which is the Severn situation.

So the figures here **must not be compared to, or converted into, a DNL 65 figure**, and must not be fed into DNL-based annoyance or reading-score formulas. The model answers one question: *how loud is each pass, where.*

INPUTS

What goes in (all measured or from public records)

Input	Source	Description
Departure tracks	ADS-B track database	90 clean 15R departures after filtering — the measured climb and west-turn geometry.
Turn-altitude record	153 daily noise logs	~30,000 departure samples → 2,889 clean 15R overflights with altitude at closest approach.

Input	Source	Description
Corridor homes	Anne Arundel County address points	Residential addresses, swept to the 62 dBA peak corridor of either track → 33,206 homes.
Noise event logs	IEC 61672 Class 2 monitor	Per-event dBA, slant range, altitude — used to fit and validate the law.

No modeled manufacturer noise tables (ANP/AEDT) are used; the law is fit directly from the TrueNoise measurements.

THE CORE CALCULATION

The loudness-vs-distance law — in plain terms, then exactly

In plain terms. Sound gets quieter with distance in a regular way, measured directly: **every halving of the distance to the aircraft adds about 7 dB, and every doubling subtracts about 7 dB.** Anchor that to one measured point — about **73 dBA when the plane's closest approach is 0.55 mi away** — and the loudness at any distance follows:

Closest approach (straight-line)	Modeled peak	Everyday comparison
0.25 mi	~81 dBA	Garbage disposal; diesel truck passing close
0.55 mi	~73 dBA	The project calibration point (measured 73.3)
1.0 mi	~66.5 dBA	Busy traffic from the sidewalk
2.0 mi	~59 dBA	Normal conversation
4.0 mi	~52 dBA	Light rain; a quiet street

Exactly. Peak falls with slant range (the 3-D straight-line distance from home to aircraft at closest approach — horizontal offset and altitude combined):

```
peak_dBA = A + B × log10(slant_range_mi)
A = 66.54   B = -24.37
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A (66.54) is the intercept — the modeled peak at exactly one mile, where the log term is zero. It is the anchor of the curve, not a threshold. **B (−24.37)** is the slope — how fast

loudness falls with distance (the "~7 dB per halving" above). Slant range is floored at 0.10 mi so the log stays finite directly beneath the path. The law was fit by regressing the single loudest sample of each of **249 overflights** against log-distance.

Validation — three independent checks

(1) An untouched calibration point. The fitted law lands at 72.9 dBA at 0.55 mi against the project's measured 73.3 — 0.4 dB, not tuned.

(2) End-to-end, against the monitor's own exposure. With the measured climb in place, the full model reproduces the receptor's level: **70.6 dBA modeled vs 70.5 measured** (median of 249 logged overflights there). The climb and geometry were not tuned to this — it validates the climb curve and the law jointly, and it is what exposed and fixed an earlier contaminated climb.

(3) A live, blind test against a calibrated meter — next.

VALIDATION · 5 FLIGHTS · BLIND

Live cross-check against a calibrated meter

On 15 October 2026, five consecutive 15R departures over a home ~130 ft from the monitor were measured three ways at once: the **model** (from the ADS-B slant range, computed before the logs were opened), a commercial **airnoise.io** device, and the **calibrated IEC 61672 Class 2 meter** (ground truth).

#	Aircraft	Slant (mi)	Model	airnoise	Calibrated	Model−cal
1	737-800	0.58	72.3	71.3	72.8	−0.5
2	A320	0.85	68.2	66.5	67.4	+0.8
3	737-800	0.83	68.5	67.0	67.3	+1.2
4	737-800	0.66	71.0	68.9	70.2	+0.8
5	A300F	0.97	66.8	66.7	66.6	+0.2

Against the meter the **model ran +0.50 dB mean, 0.76 dB RMS, 1.2 dB worst case**; the commercial device −0.78 dB mean, 0.98 dB RMS. Both landed well inside the ±7 dB

screening band, the model slightly ahead. *Caveat:* n = 5, one 10-minute window, one weather state, no near-overhead pass — a confirmation, not a second calibration.

MEASURED ALTITUDE · 2,889 FLIGHTS

How high the aircraft actually are

The most-contested point in community discussion is how high aircraft are when they turn. It has a measured answer on a large sample, and the answer is that **there is no single turn altitude** — the turn is a process, not a point.

Across **2,889 clean 15R departures**, altitude at closest approach has a **median of 2,325 ft** (middle half 2,141–2,617; only 7.8% below 2,000 ft). Closest approach falls a median 3.46 nm out — the turn-completion zone.

Putting the geometry together: departures **begin the turn around 1,800 ft** (near 2 nm), **complete it around 2,300 ft** (2.5–3 nm out, over the community), and keep climbing west. They are **lowest over the close-in neighborhoods and schools** nearest the runway and higher by the western end. Quoting any one height for "the turn" is what fuels the confusion.

Month (2026)	Median completion altitude
May	2,325 ft
June	2,375 ft
July	2,292 ft
August	2,196 ft

This also sets the stakes for proposals that turn aircraft lower: on the measured falloff, dropping the turn from ~2,300 ft to ~1,000 ft adds on the order of 6 dB to the peak directly beneath it.

Altitude convention. These figures use the logging convention in place through 2026-10-19. From 2026-10-20 the measurement app references altitude to mean sea level (GNSS geoid / barometric-QNH correction), which reads about 100 ft higher for the same aircraft; figures recomputed on new-convention data will sit correspondingly higher, and old and new logs must not be pooled. The new datum (WGS-84 ellipsoid) is pending confirmation against a live reference-track measurement.

MODELING CHOICES

The decisions that define the model

- **Metric = single-event peak dBA.** DNL and $L_{Aeq,16h}$ are deliberately excluded.
- **Both procedures as idealized single lines.** WARYN-BRGHT is notional, so it has no measured spread; for a fair comparison today's 15R is also its median centerline, not its cloud.
- **Today's line = the ~79% west-curve majority**, built along-track and extended to the

published BOBYJ leg.

- **WARYN-BRGHT line** shares the runway segment (so near-in differences are ~0), then runs WARYN → BRGHT.
- **Altitude = the measured 15R climb**, draped by along-track distance — the piece independently confirmed by the 2,889-flight record and the receptor check.
- **Peak = the law at the 3-D minimum slant range** between each home and the line.

UNCERTAINTY

How precise the numbers are

Error band = ±7 dB (95%). The residual standard deviation of the 249-point fit is 3.7 dB, reflecting real variation across aircraft type, weight and weather. **Differences smaller than this band are not individually distinguishable per home.** The defensible reading is the *spatial pattern* of where exposure shifts, not any single home's exact value — except the receptor's own level, which is anchored to measurement. (Modeling aircraft type explicitly tightens the band to about ±6.3 dB; that refinement is in progress and is not used in the map figures.)

RESULTS

What the model finds

Corridor (33,206 homes), single-event peak dBA — counts by band (<55 / 55–60 / 60–65 / 65–70 / ≥70):

Scenario	<55	55–60	60–65	65–70	≥70
Today's 15R	879	3,492	10,019	14,512	4,304
WARYN-BRGHT	4,501	5,317	11,627	7,210	4,551

By per-home change: quieter by ≥1.5 dB **16,772**, within ±1.5 dB **7,256**, louder by ≥1.5 dB **9,178**; **median change −1.75 dB** — net relief by roughly two-to-one, at the cost of a real

increase on the southeast side.

Schools (peak dBA, ±7 dB), most-exposed-today first:

School	Today	WARYN-BRGHT	Δ
Quarterfield Elem.	74.5	70.9	−3.6
Glen Burnie Park Elem.	70.2	71.3	+1.2
Rippling Woods Elem.	64.3	68.8	+4.5
Severn Elem.	62.1	58.3	−3.7
Southgate Elem.	59.9	63.1	+3.2

The most-exposed school today, Quarterfield, is among those helped most; two schools (Rippling Woods, Southgate) see real increases, stated plainly. The robust, cross-option fact is about altitude over children: **WARYN-BRGHT is the only option studied that never sends a pass below ~2,000 ft over any school, while a low turn drops a ~1,800-ft overflight onto the schools nearest the runway.**

CLEARING IT UP

Common misconceptions

“TrueNoise's numbers are above the FAA's 65 limit.”

No — these are single-event peaks (the loudest instant of one flyover); 65 is a DNL 24-hour-average land-use contour. Different quantities, different scales. Our numbers cannot be compared to, or converted into, a DNL 65 figure.

“What altitude do they turn at?” — there isn't one number.

On 2,889 measured flights, the turn begins around 1,800 ft and completes around 2,300 ft over the community, climbing west. Aircraft are lowest over the close-in neighborhoods and schools. Quoting one height — high or low — is what fuels the confusion.

“The monitor's readings are the worst case.”

The opposite. The fixed monitor sits ~3.4 nm out, in the turn-completion zone, where its measured median peak is 70.5 dBA. Homes closer to the turn, where aircraft are lower, hear louder passes. The published receptor figures are a conservative, low-end anchor, not a maximum.

“Spreading the flights out will fix it.”

Spreading moves noise; it does not lower it. Loudness is set by altitude, not by how tightly the tracks bunch. Fanning a low turn across more streets spreads a louder-per-event overflight onto more homes — and if the turn is forced lower to do it, every home under the wider band is louder. Only raising the turn makes each pass quieter for everyone beneath it.

INSTRUMENTS

Equipment and its limits

Measurements come from an **IEC 61672 Class 2** monitor and a co-located airnoise.io device. Class 2 is the right grade here for two reasons. First, the analysis is **comparative** — today's procedure against a proposed one at the same homes — so any fixed instrument offset applies to both and cancels in the difference. Second, the instrument has been checked against a **calibrated meter** and found within ~0.8 dB, well inside the ±7 dB screening band. This is a measured community screening, **not a certified Class 1 network and not an FAA AEDT model**; where a regulatory-grade number is required, these figures show *where* to look, not the final certified value. The measurement app’s full signal chain — A-weighting,

SPL computation, calibration, psychoacoustic analysis, and ADS-B correlation — is documented in the Aircraft Noise Tracker technical brief.

CARRY THESE WITH ANY USE

Scope limits and honest caveats

- **Metric:** peak dBA only — not DNL, not $L_{Aeq,16h}$. Do not convert to either.
- **The west-curve share is load-bearing.** The whole west-relief / southeast-increase split rests on ~79% of today's departures curving west; that share is measured but not yet ground-truthed against a full operations record.
- **The single line is weakest where flights fan out** — between the turn and BOBYJ the measured tracks diverge.
- **The climb is a median;** individual departures run higher or lower (part of the ±7 dB band).
- **Idealized single lines** understate how many receptors real dispersed operations affect; precise dispersion is for FAA AEDT modeling.
- **WARYN-BRGHT is unflown** — projection, not observation.

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